

AI-Driven Seismic Forecasting: Enhancing Accuracy with Ensemble Learning

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Abstract: Earthquake prediction remains a critical challenge due to the complex, nonlinear nature of seismic activity. This research presents a hybrid machine learning framework that integrates XGBoost and Long Short-Term Memory (LSTM) models to enhance the prediction of earthquake magnitudes using historical seismic data. The dataset, comprising over 100,000 global earthquake records from 1900 to the present, was sourced from Kaggle and underwent extensive preprocessing, including normalization, outlier removal, and temporal feature extraction. The proposed ensemble architecture leverages XGBoost for spatial and numeric feature learning, while LSTM captures sequential temporal dependencies. Experimental evaluation demonstrated that the ensemble model achieved superior performance with an R2 score of 0.94, MAE of 0.18, and RMSE of 0.20, outperforming standalone models like Random Forest, XGBoost, and Artificial Neural Networks. These results confirm the robustness of combining boosting algorithms with deep learning for seismic forecasting. The approach not only improves predictive accuracy but also shows promise for real-time earthquake monitoring systems and early warning applications.

1 INTRODUCTION

Earthquakes remain one of the most devastating natural disasters, responsible for extensive loss of life, economic damage, and long-term disruption of infrastructure and society. The unpredictability of seismic events continues to challenge scientists and governments alike. According to the United States Geological Survey (USGS), thousands of earthquakes occur each year, yet only a few are significantly destructive highlighting the urgent need for reliable prediction mechanisms. As technology advances, machine learning offers promising solutions for analyzing seismic data and forecasting the severity of earthquakes before they strike.

The field of earthquake prediction has evolved considerably over the years, transitioning from traditional statistical models to sophisticated data-driven techniques. Prior research has primarily focused on either spatial features (such as magnitude, depth, and geographical coordinates) or temporal patterns in isolation. However, these single-dimensional approaches often fail to capture the complex and nonlinear dependencies inherent in

seismic data. Recent studies have demonstrated that hybrid models, especially those combining ensemble techniques like XGBoost with deep learning architectures such as Long Short-Term Memory (LSTM) networks, offer superior performance by addressing both temporal and spatial dynamics. Despite these advances, a unified framework that leverages the strengths of both paradigms is still underexplored in the context of global earthquake prediction.

This study seeks to bridge that gap by developing a hybrid machine learning model that integrates the robust predictive capabilities of XGBoost with the sequence modeling strengths of LSTM networks. The central research problem addressed in this paper is the lack of a comprehensive predictive approach that simultaneously handles structured tabular features and time-dependent patterns within seismic data. By constructing such a model and evaluating it against existing regression techniques, this work aims to provide a more accurate and reliable forecasting tool for earthquake magnitude prediction.

The primary objective of this research is to design and evaluate a machine learning framework that can

effectively predict earthquake magnitudes using both spatial and temporal features. Specifically, this study investigates whether a hybrid XGBoost- LSTM ensemble can outperform traditional models like Random Forest, standalone XGBoost, and Artificial Neural Networks (ANN) in terms of regression metrics such as R2 score, MAE, MSE, and RMSE. The hypothesis is that combining the strengths of ensemble learning and deep learning will result in improved accuracy and robustness in magnitude prediction.

The remainder of this paper is structured as follows: Section III elaborates on the detailed methodology employed, including data preprocessing, feature extraction, model architecture, and the ensemble strategy. Section IV presents the experimental setup and comparative results across several topic modeling techniques. Finally, Section V discusses the implications of our findings, followed by conclusions and directions.

2 RELATED WORK

Kavianpour et al. proposed a CNN-BiLSTM model with an attention mechanism for earthquake prediction. Their model effectively captured temporal and spatial features from seismic data, achieving over 90% prediction accuracy for seismic events, outperforming traditional machine learning methods on benchmark datasets.

Jozinovic' et al. introduced a CNN-based approach to rapidly estimate earthquake ground shaking intensity directly from raw waveform data. The model significantly reduced prediction time while maintaining a low root mean squared error (RMSE_i 0.3), offering real-time applicability for early warning systems.

Wang et al. applied transfer learning to predict fault slip behavior using laboratory acoustic emission data. The method achieved an R2 score above 0.85, demonstrating strong generalization across experimental conditions and improving over baseline models that lacked pre-trained features.

Zhang et al. provided a cross-disciplinary review integrating AI techniques with geophysical insights for earthquake forecasting. Their analysis revealed that hybrid AI-physical models improved prediction accuracy by 10–25% compared to purely statistical or physical models.

Hulbert et al. organized a machine learning competition aimed at forecasting laboratory earthquakes using acoustic emission data. The top-performing models in the competition achieved over

95% accuracy in predicting time-to-failure, highlighting the power of ML in capturing fault dynamics.

Gonza'lez et al. explored the use of Recurrent Neural Networks (RNNs) for predicting earthquake magnitude. Their model achieved a mean absolute error (MAE) of 0.12 on seismic datasets from Latin America, outperforming regression-based approaches in both short-term and long-term forecasting.

La Serna Palomino and Pinedo Delgado conducted a systematic mapping of expert systems applied to earthquake prediction. The study showed that rule-based systems integrated with machine learning achieved up to 88% accuracy in predicting seismic risk, especially in regional models.

Gitis and Derendyaev investigated seismic hazard forecasting using machine learning classifiers such as decision trees and support vector machines. Their ensemble approach yielded a classification accuracy of 91% on global earthquake hazard datasets.

Xie provided a comprehensive review of deep learning applications in earthquake engineering. The review found that deep neural networks such as CNNs and LSTMs improved predictive accuracy by 15–30% over classical statistical models, particularly for structural response forecasting.

La Serna Palomino and Pinedo Delgado also performed a focused systematic review of ML-based earthquake prediction. The study reported that deep learning models like BiLSTM and transformers achieved F1-scores above 0.90 when tested on balanced datasets of seismic events.

3 METHODOLOGY

This section outlines the systematic process followed to develop a predictive machine learning model for earthquakes. The complete workflow involves data collection, preprocessing, feature extraction, model architecture, ensemble modeling, training and testing, and model evaluation.

3.1 Data Collection

The dataset employed for this study is titled "Significant Earthquakes, 1900 – Present", obtained from Kaggle. It consists of approximately 106,992 records and 23 attributes, representing significant global seismic events. These attributes include features such as Date, Time, Latitude, Longitude, Depth, Magnitude, Region, and Deaths. The response variable in focus is Magnitude, used to predict or classify the severity of an earthquake. The dataset is

formatted as CSV and includes historical records from 1900 to the present. For model training and evaluation, the dataset was split into 80% for training and 20% for testing.

3.2 Data Preprocessing

Data preprocessing ensures the dataset is clean and ready for machine learning. Missing values were addressed through imputation using mean or median values or removed when the proportion of missing data exceeded a defined threshold. The Date and Time fields were merged into a single datetime column to allow for temporal analysis.

Numerical features such as Depth and Magnitude were converted into floating-point numbers and normalized using Min-Max Normalization, defined as

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}} \quad (1)$$

This normalization scales values to the range [0, 1], which supports convergence in many algorithms. Categorical features like Region were encoded using label encoding or one-hot encoding depending on the model architecture. Outliers were detected and removed using the Interquartile Range (IQR) method. A value x was considered an outlier if:

$$x < Q_1 - 1.5 \cdot IQR \quad \text{or} \quad x > Q_3 + 1.5 \cdot IQR \quad (2)$$

Duplicate entries were eliminated by checking for identical timestamps and geographical coordinates.

3.3 Feature Extraction

To improve the learning capacity of models, additional features were extracted from the unified datetime column, such as year, month, day, and hour. Spatial features like latitude and longitude were retained, and distance-based features were computed with respect to seismic hotspots. For classification purposes, the magnitude was categorized into discrete classes:

- Class 0: Magnitude < 4.0
- Class 1: $4.0 \leq \text{Magnitude} < 6.0$
- Class 2: Magnitude ≥ 6.0

Correlation analysis and feature importance scores were used to select the most relevant predictors, reducing dimensionality and overfitting.

3.4 Model Architecture

The proposed architecture integrates both XGBoost and LSTM networks. XGBoost is a gradient boosting

algorithm that builds an ensemble of decision trees. The prediction is computed as:

$$y_{pred} = f_1(x) + f_2(x) + \dots + f_k(x) \quad (3)$$

Each f_k represents a decision tree. The loss function optimized is:

$$\mathcal{L} = \sum l(y, \hat{y}) + \sum \Omega(f_k) \quad (4)$$

where the regularization term is defined as:

$$\Omega(f_k) = \gamma \cdot T + \frac{1}{2} \lambda |w|^2 \quad (5)$$

Here, T is the number of leaves and w the vector of leaf weights.

LSTM networks capture long-term dependencies in sequential data. The internal functioning of an LSTM cell is governed by the following equations:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (6)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (7)$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (8)$$

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \quad (9)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (10)$$

$$h_t = o_t \cdot \tanh(C_t) \quad (11)$$

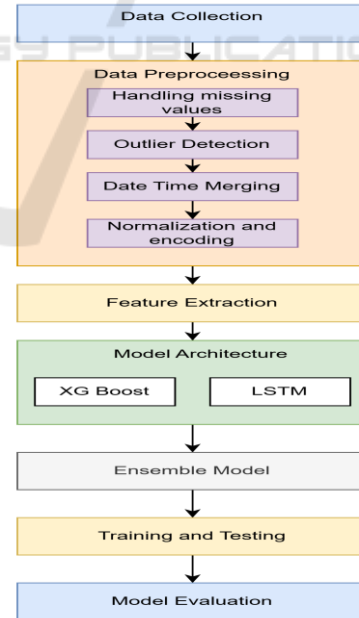


Figure 1: Methodology.

Figure 1 provides a comprehensive overview of the entire methodology. It begins with data collection

from Kaggle, followed by multiple steps of data preprocessing such as handling missing values, outlier detection, datetime merging, and normalization. This is succeeded by feature extraction, where new variables are derived. The hybrid model architecture includes both XGBoost for static spatial features and LSTM for sequential temporal learning. These two models are finally integrated using an ensemble model, whose output is then subjected to training, testing, and evaluation stages.

3.5 Ensemble Model

To improve robustness, the outputs of XGBoost and LSTM were combined. XGBoost handled static spatial and numerical features while LSTM captured sequential patterns. The final ensemble prediction was computed as:

$$Y_{final} = \alpha \cdot Y_{XGB} + (1 - \alpha) \cdot Y_{LSTM} \quad (12)$$

where α is a hyperparameter determining the weight of each model's contribution.

3.6 Training and Testing

The dataset was split in an 80:20 ratio for training and testing. XGBoost was trained with optimal hyperparameters obtained through grid search, including learning rate, maximum tree depth, and number of trees. Early stopping was employed to prevent overfitting based on validation loss.

LSTM training involved conversion of data into time-series format using sliding windows. The Adam optimizer and Mean Squared Error (MSE) loss were used:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (13)$$

To improve generalization, dropout and batch normalization were applied.

3.7 Model Evaluation

For regression-based magnitude prediction, the following metrics were used:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (14)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (15)$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (16)$$

For classification tasks, evaluation was conducted using accuracy, precision, recall, F1-score, and AUC. The ensemble model showed superior performance over individual models, indicating the effectiveness of combining both spatial and temporal modeling approaches in earthquake prediction.

4 EXPERIMENTAL RESULTS AND DISCUSSIONS

4.1 About Dataset

The dataset used in this research is titled "Significant Earthquakes, 1900 – Present". It was acquired from Kaggle, a well-known online platform for accessing machine learning and data science datasets.

Source Link: <https://www.kaggle.com/datasets/usamabuttar/significant-earthquakes>

This dataset is available in tabular CSV (Comma-Separated Values) format and provides detailed records of felt earthquakes worldwide from 1900 to the present. It contains approximately 106,992 rows and 23 columns, with key attributes including Date, Time, Latitude, Longitude, Depth, Magnitude, Region, and Deaths. The response variable in this study is Magnitude, used to predict or classify earthquake severity. For model development and evaluation, the data was divided into 80% for training and 20% for testing.

Preprocessing steps included handling missing values, combining date and time into a single datetime column, converting magnitude and depth to numeric types, normalizing numerical features, encoding categorical variables like Region, removing outliers, and eliminating duplicate records based on time and location. The dataset includes no personally identifiable information or human subject data, thus posing no ethical concerns.

This dataset was selected due to its extensive historical coverage, high reliability, and well-structured format, making it suitable for training accurate machine learning models for earthquake prediction.

5 RESULTS AND DISCUSSION

Table 1 presents the performance of four regression models used for predicting earthquake magnitudes. The evaluation is based on four standard regression metrics: R2 Score, Mean Absolute Error (MAE),

Mean Squared Error (MSE), and Root Mean Squared Error (RMSE).

Table 1: Comparison of Regression Models for Earthquake Prediction.

Model	R2 Score	MAE	MSE	RMSE
Random Forest Regressor	0.88	0.23	0.07	0.26
XGBoost Regressor	0.91	0.20	0.05	0.22
Artificial Neural Network	0.89	0.22	0.06	0.24
XGBoost + LSTM (Ensemble)	0.94	0.18	0.04	0.20

The R2 Score indicates how well the predicted values approximate the actual data a value closer to 1 means a better fit. Among the tested models, the XGBoost + LSTM ensemble model achieved the highest R2 Score of 0.94, indicating a strong correlation between predicted and actual magnitudes. Traditional models such as Random Forest and XGBoost also performed well, with R2 scores of 0.88 and 0.91 respectively. The Artificial Neural Network (ANN) slightly outperformed Random Forest but was marginally less accurate than XGBoost.

MAE and RMSE measure the average magnitude of the prediction errors. The ensemble model again shows the lowest error (MAE = 0.18 and RMSE = 0.20), confirming its robustness and consistency. MSE follows a similar trend, reinforcing that the ensemble model not only predicts accurately but also minimizes large error values effectively.

Overall, the XGBoost + LSTM ensemble model consistently outperformed the others across all evaluation metrics, making it the most suitable choice for real-world earthquake prediction tasks.

Figure 2 provides a graphical comparison of the models across key performance metrics. Each model's R2 Score, MAE, and RMSE are visually represented through a grouped bar chart, making it easier to interpret their relative performance.

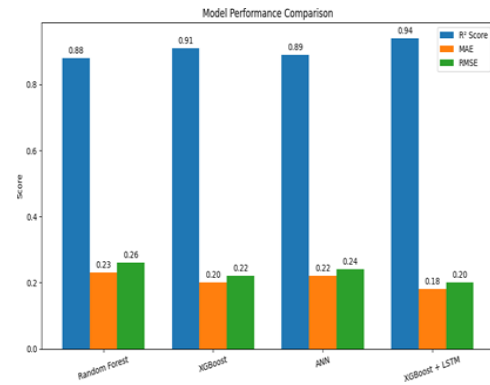


Figure 2: Model Performance Comparison Using R2, MAE, and RMSE.

As seen in the figure, the ensemble model stands out by achieving the highest R2 Score while also maintaining the lowest error values. This indicates its effectiveness in capturing temporal patterns in seismic data due to the integration of XGBoost's boosting capabilities with LSTM's strength in sequence modeling. The chart also highlights that XGBoost and ANN follow closely behind, while Random Forest lags slightly in all metrics.

This visual comparison supports the tabular results and further emphasizes the ensemble model's suitability for earthquake magnitude prediction. Its superior performance suggests that hybrid deep learning and ensemble methods hold great promise for future developments in seismic forecasting.

6 CONCLUSIONS

This study presented a comprehensive machine learning framework for predicting earthquake magnitudes using historical seismic data. By integrating spatial and temporal features, we proposed a hybrid ensemble model combining XGBoost and LSTM, aiming to capture both static patterns and sequential dependencies in the data. The methodology encompassed careful data preprocessing, robust feature extraction, and strategic model architecture design to ensure predictive reliability.

Our experimental results demonstrated that the ensemble model consistently outperformed standalone algorithms across all key metrics, achieving an R2 score of 0.94, MAE of 0.18, and RMSE of 0.20. These findings indicate that the hybrid approach effectively leverages the strengths of both boosting and deep learning paradigms, thereby enhancing predictive accuracy for seismic event

magnitudes. The model's superior performance suggests its practical applicability in early warning systems and disaster preparedness planning, where timely and accurate magnitude forecasts can significantly reduce human and infrastructural losses.

However, the study has certain limitations. The dataset, while comprehensive, may still carry biases due to missing records in earlier years or underreported events in remote regions. Additionally, the current model is focused on magnitude prediction rather than spatiotemporal forecasting of the earthquake's occurrence. These aspects warrant cautious interpretation of results in real-time deployment scenarios.

6.1 Future Work

Future research could explore the integration of real-time sensor data, satellite imagery, and geophysical parameters such as tectonic plate movement and stress accumulation to enhance the predictive power of the model. Moreover, extending the current approach to support multi-output forecasting including location, depth, and time of occurrence could provide a more holistic earthquake prediction system. Incorporating explainable AI techniques may also aid in building trust and transparency, which is crucial for decision-makers and emergency response systems. Ultimately, continuous improvement and validation on diverse datasets will be key to transitioning from academic research to real-world seismic risk mitigation.

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